

Checkpoint 4: Residual Denoising Backbone

Objective

Replace the U-Net denoising backbone with a ResNet-style residual backbone under the same diffusion task.

Key Message

The proposed residual backbone improves structural and color-oriented metrics over the default baseline.

Completed Work

- Implemented residual blocks and residual denoising backbone.
- Trained the residual model using the same paired split and objective.
- Compared residual and baseline outputs across three seeds on the same 384 test images.

Verification and Controls

- Reused the same manifests, optimizer family, diffusion objective, evaluation code, and test tensors.
- Stored each seed as an independent run before calculating model-level mean and sample standard deviation.
- Checked structural and color metrics separately from pixel-error metrics to avoid a one-number conclusion.

Evidence

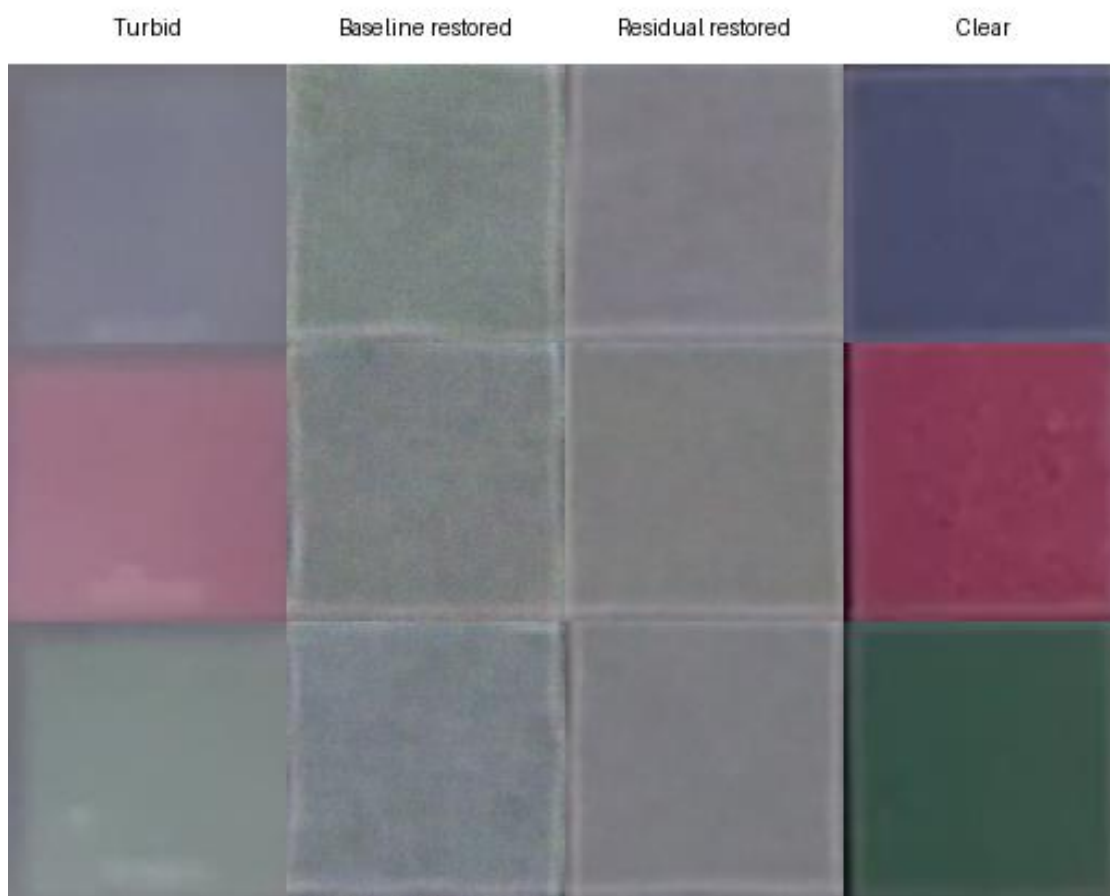
Metric	U-Net mean +/- SD	Residual mean +/- SD
MSE	0.039873 +/- 0.001756	0.038782 +/- 0.003119
PSNR	15.780514 +/- 0.187075	16.006089 +/- 0.267086
SSIM	0.657180 +/- 0.015721	0.779948 +/- 0.006607
Delta E	28.380208 +/- 1.320879	27.053617 +/- 1.081249

Decision / Next Gate

Advance the residual backbone to capacity-controlled comparison while qualifying topology and parameter-count differences.

Notes

- Residual has better mean SSIM and Delta E than the default U-Net; topology, depth, and capacity differ together.



Matching default U-Net and residual restoration examples.